Exercise 3 (12p)

A biometric system is working in VERIFICATION mode. We carry out different genuine and impostor tests. In concrete, 1750 matchings of samples from the same person (genuine) and 1900 matchings of samples from different persons (impostor) are done. In each matching, we employ a DISTANCE measure, so the LOWER the value of the measure, the MORE SIMILAR are the samples.

Results of the experiments are reported in the following histograms. The first histogram is when samples from the same person are matched (genuine), and the second histogram is when samples from different persons are matched (impostor). The x-axis of the histogram indicates the distance value, and the y-axis indicates the number of matchings. A table with results from the histograms is also reported.

Questions

a) Calculate the FAR and FRR of the system for different thresholds of the distance measure. Report you results both in:

-a table, indicating the threshold and the respective errors (employ at least 10 different thresholds, with at least 6 different thresholds where the two histograms overlap)

-a graph (draw errors in the range 0-30% approximately), indicating clearly in the axes the values of each point

Value: showing the table: 5p, showing the graph 4p (total 9p)

b) Find the approximate threshold of the EER (Equal Error Rate) and the value of the EER in your graph of a).

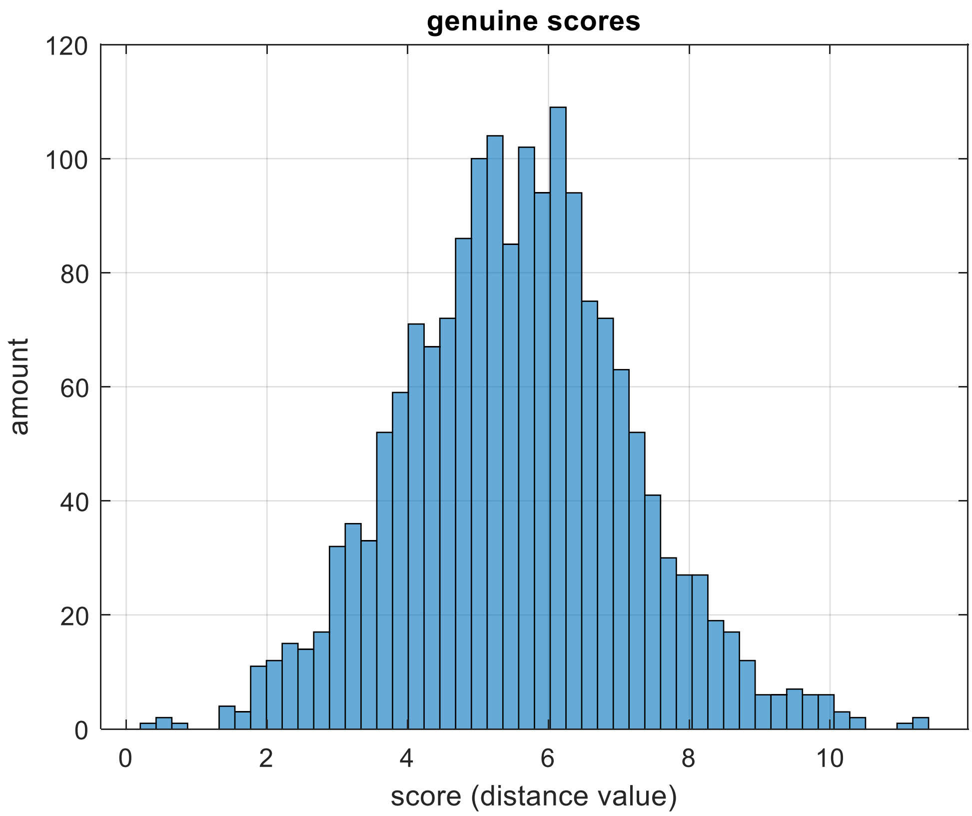
Do not forget to mark the values in your graph of a)!!

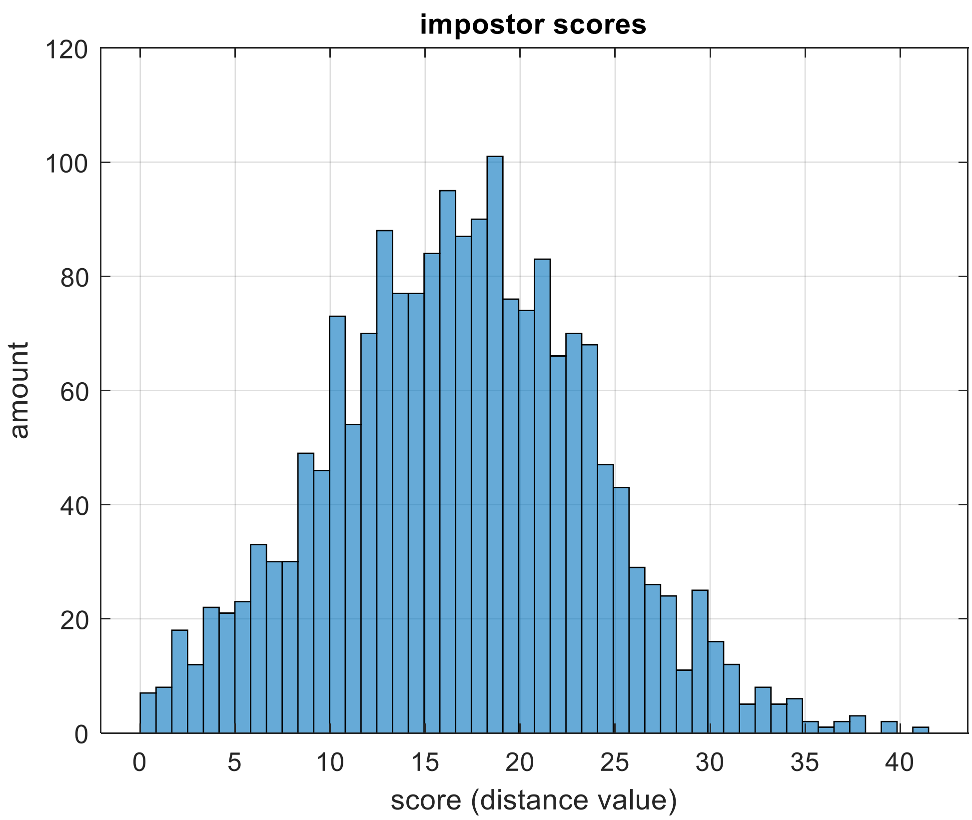
Value: 1p

c) Find the approximate threshold in your graph of a) at which FRR (False Rejection Rate) is about 3% and indicate the value of FAR (in %) at this threshold.

Do not forget to mark the values in your graph of a)!!

Value: 2p


genuine scores
amount
score (distance value)
0
20
40
60
80
100
120
0
2
4
6
8
10


impostor scores
amount
score (distance value)
0
20
40
60
80
100
120
0
5
10
15
20
25
30
35
40

amount genuines	distance genuines		amount impostors	distance impostors
1	0.312		7	0.415
2	0.536		8	1.245
1	0.76		18	2.075
0	0.984		12	2.905
0	1.208		22	3.735
4	1.432		21	4.565
3	1.656		23	5.395
11	1.88		33	6.225
12	2.104		30	7.055
15	2.328		30	7.885
14	2.552		49	8.715
17	2.776		46	9.545
32	3		73	10.375
36	3.224		54	11.205
33	3.448		70	12.035
52	3.672		88	12.865
59	3.896		77	13.695
71	4.12		77	14.525
67	4.344		84	15.355
72	4.568		95	16.185
86	4.792		87	17.015
100	5.016		90	17.845
104	5.24		101	18.675
85	5.464		76	19.505
102	5.688		74	20.335
94	5.912		83	21.165
109	6.136		66	21.995
94	6.36		70	22.825
75	6.584		68	23.655
72	6.808		47	24.485
63	7.032		43	25.315
52	7.256		29	26.145
41	7.48		26	26.975
30	7.704		24	27.805
27	7.928		11	28.635
27	8.152		25	29.465
19	8.376		16	30.295
17	8.6		12	31.125
12	8.824		5	31.955
6	9.048		8	32.785
6	9.272		5	33.615
7	9.496		6	34.445
6	9.72		2	35.275
6	9.944		1	36.105
3	10.168		2	36.935
2	10.392		3	37.765
0	10.616		0	38.595
0	10.84		2	39.425
1	11.064		0	40.255
2	11.288		1	41.085